

Report conversation

if testing a code requires unit testing private methods,
what does that say about the design?

Testing private methods often indicates that the design of the class or module may not be optimal. Private methods are intended to be internal implementation details, not exposed for external testing. If you find a need to test private methods directly, it could suggest a few things:

1. **Complexity:** The private method might be too complex or performing too many tasks, indicating a potential violation of the Single Responsibility Principle. It could be a sign that this method should be refactored into a separate class or module, which can then be tested independently.
2. **Cohesion:** The class or module may not be cohesive enough, meaning it's doing more than it should. A well-designed class should have a clear, single purpose, with private methods closely related to the class's main responsibility.
3. **Abstraction:** The design may not be abstracting the right set of behaviors. If a private method is crucial enough to require direct testing, it might represent a significant behavior that should be part of a public API, either of the current class or another class.
4. **Testability:** The overall testability of the code might be low. Good design often correlates with testability, and private methods that require testing can signal a need to rethink how the code is structured.

In summary, while there are occasional justifications for testing private methods, frequently feeling the need to do so should prompt a review of the design for better abstraction, cohesion, and adherence to principles like Single Responsibility and Open/Closed.